

Analytical Assessment

Let the Universe of Discourse be the set of all running processes on a Linux system.

Let $Z(x)$ be the predicate: "Process x is a zombie process."

Let $R(x)$ be the predicate: "Process x is running as root."

- Analyze and answer the following:

1- Translate the following English statement into formal logic:
"There does not exist a process that is a zombie process."

2- Use the laws of negation to find the logical equivalence of your answer in question 1, and write the new formal logic expression.

3- Translate the new formal logic expression from question 2 back into a clear English sentence.

4- Negate the formal statement $\forall x R(x)$ and translate the resulting negated expression into an English sentence.

Solution:

$$1) \neg(\exists(x) Z(x))$$

$$2) \forall(x) \neg Z(x) \equiv \neg(\exists(x) Z(x))$$

3) For all processes x , x is not a zombie processes

4) There exists at least one process that is not running as root